

PAPER PRESENTATION - STUDENT HANDOUT

BIOSTATISTICS 755, SPRING 2026

Purpose

The purpose of this assignment is to develop your ability to find, explain, and critically evaluate a peer-reviewed public-health research article that uses longitudinal or clustered data. By completing this assignment, you will gain experience interpreting applied longitudinal analyses, assessing whether the chosen methods appropriately address the scientific question, and clearly communicating statistical ideas to an applied audience. Emphasis should be placed on understanding the research question, study design, modeling strategy, and interpretation of results rather than on statistical theory.

What Papers Are Eligible

The selected paper must:

- Be published in a peer-reviewed journal.
- Address a public health, epidemiologic, clinical, or health services research question.
- Use longitudinal or clustered data (e.g., repeated measures, panel data, multilevel data).
- Apply a statistical method appropriate for correlated data (e.g., mixed-effects models, GEE, multilevel models, or related approaches).

Papers that rely solely on cross-sectional analyses are not eligible. Methodological papers are generally discouraged unless they include a substantial applied data example that is the primary focus of your presentation.

Deliverables

Proposal (Due February 2)

You will submit a brief proposal that includes:

- The full citation of the paper and a working link to the article.
- A short justification (no more than 250 words) describing:
 - The research question and study context,
 - Why the paper is of interest to you, and
 - Why longitudinal or clustered data methods were necessary for the analysis.

Slides

Presentation slides (approximately 10–12 slides) must be emailed to the instructor at least 24 hours before your scheduled presentation date. Please post a link to your paper in slack 3 days before your presentation.

Presentation

Each student will give a ~20-minute in-class presentation. All presentations will be individual (no group presentations). Presentations will begin on February 10 and continue through the middle of April.

Suggested Presentation Structure

While exact organization is flexible, a typical presentation might include:

- Brief background and motivation for the study
- Research question(s) and study design
- Description of the data and longitudinal or clustered structure
- Statistical methods used, with emphasis on:
 - Why the chosen statistical methods (e.g., linear mixed model, generalized estimating equations) were chosen
 - How correlation or clustering was handled
 - How the covariate structure (e.g., group, linear time effect, and a group by linear time interaction) answers the question(s) of interest
 - Alternative structures to answer the same question
- Key results and scientific conclusions
- Critical evaluation:
 - Strengths and limitations of the analysis
 - Appropriateness of the modeling choices
 - Any alternative approaches you might consider

The goal is not to reproduce every technical detail, but to demonstrate clear understanding of the analysis and its implications.

Logistics and Tips

A sign-up link will be circulated during the semester for you to select a presentation date. Dates are first-come, first-served, and everyone is expected to present on their chosen day.

Additional tips:

- Assume your audience has taken Biostatistics 755 but has not read the paper.
- Focus on interpretation and reasoning rather than mathematical details.
- Clearly explain how the longitudinal or clustered structure influenced the analysis.
- Practice timing to ensure you stay within the allotted 20 minutes.

Grading Rubric

Component	Points
Proposal quality and appropriateness of paper	15
Understanding of study design and data structure	20
Explanation of statistical methods	25
Interpretation of results and conclusions	20
Critical evaluation and discussion	15
Clarity, organization, and delivery	5
Total	100